8.0 Uncovering the Gems of the Blockchain: Meet the Miners and the Mining Process

Miners are the bookkeepers.

- In centralized systems, accountants get paid by companies to keep track and maintain the accuracy and integrity of their financial records.

Similarly, **miners** are paid in ***bitcoin*** for their work in verifying and adding transactions to the *blockchain*, helping to keep the network secure and running smoothly. This job involves the use of **computational power** and specialized hardware. The objective of mining is to add new blocks to the *blockchain* and maintain its security, decentralization, and long-term viability.

What is Bitcoin Mining?

Bitcoin mining is a process of adding transaction information to **Bitcoin**'s public journal of past transactions, or a blockchain.

This ledger of past transactions is called the block chain as it is a string of blocks. This chain serves to validate the transactions of all those other networks as having occurred.

Bitcoin nodes use this technology chain to distinguish a genuine **Bitcoin** transaction from efforts to re-spend coins which may have already been spent somewhere else.

Miners gather unconfirmed transactions and form a block, then embark on a search for the valuable key that will **secure the block's place in the blockchain**.

The key is a "**valid block hash**," which is hidden among billions of others and can only be unlocked by a specific key set by the network.

- Imagine a giant haystack filled with millions of keys, each representing a unique block hash. The network has set a specific key that will unlock a valuable prize. Miners search through the haystack, trying each key in the lock, but only one miner will be lucky enough to find the correct match.

The first miner to find the correct block hash broadcasts it to the network along with the block of transactions. Other miners then verify the solution to ensure it fits the lock correctly. If the solution is accurate, the block is added to the blockchain, creating a secure and public ledger.

For their hard work, miners are rewarded for their efforts in two ways: block rewards and transaction fees. **Block rewards** are newly generated ***bitcoin*** that are released into circulation with each block added to the blockchain. **Transaction fees**, on the other hand, are small amounts of ***bitcoin*** that users pay to have their transactions processed more quickly and prioritized in the block by the miner. Miners are free to choose which transactions to include in the block they mine, and they often prioritize those with the highest transaction fees.